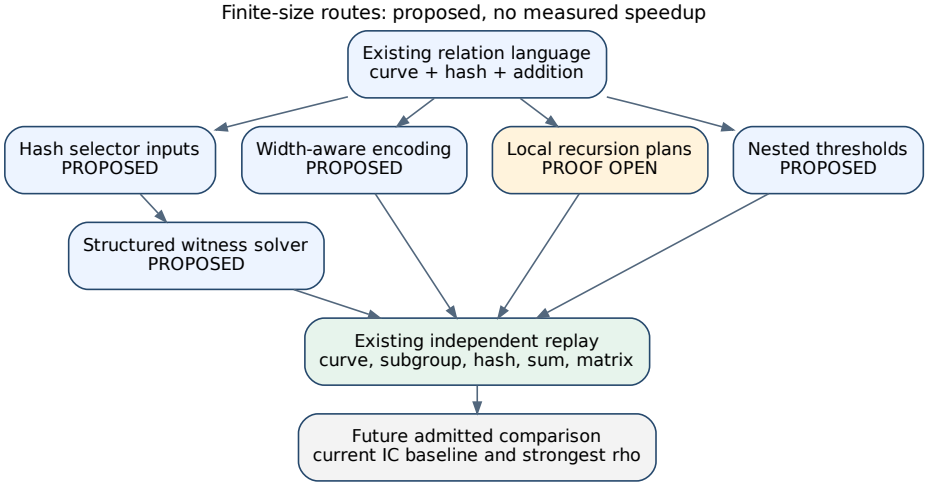
Finite-size extensions of hashed-factor-base ECDLP

Source review and proposed implementation directions

30 October 2026

Status. Planning only: no new implementation, experiment, theorem, speedup or scalar recovery. The enormous onset belongs to one general contraction schedule; it is not a lower bound for solving its particular relation networks. The original manuscript's asymptotic conclusion is conditional on its cited tensor-contraction claim, not independently validated here.



1. Hash selectors as free inputs. The native factor-base enumerator already inverts the affine hash on a bounded threshold interval. Its relation network instead allocates free point coordinates and constrains membership afterwards. A proposed search should branch on the interval selector and derive the encoded point. Roughly $\lceil \log_2 \tau \rceil$ free bits per slot replace freely chosen coordinates; auxiliary arithmetic and decoder work remain. Canonical decoding, invalid-code rejection and subgroup checks must be faithful. Fixed cofactor projection can produce multiple source assignments: assignment counts and distinct decompositions are different quantities. This reuses an existing inverse map; enumerating all selector tuples is still generic search.

2. A structured witness backend. A relation needs one verified decomposition. Existing native functional witnesses avoid repeated counts but still branch over declared free inputs. A witness-only API plus typed arithmetic constraints could support propagation, incremental target assumptions and exact SAT/SMT, including native XOR reasoning for binary fields. Opaque predicate closures are not a CNF exporter. Earlier binary SAT/enumeration encodings already have negative scaling evidence in the lead atlas. A new backend must identify useful pruning and price failures, conflicts and propagation; a wrapper alone is not an algebraic improvement. Timeouts mean UNKNOWN, never UNSAT.

3. Optimize contraction width and encoding together. Linear edge count and quick construction do not imply small intermediate tensors. Plan low-arity structured factors, elimination orders and cutsets; compare Fourier and simpler arithmetic gadgets by their complete finite-size solve cost. TensorOrder supplies a precedent for graph-decomposition-guided contraction with explicit factorability assumptions. The older binary bucket/cutset prototype already encountered large widths on tiny fields; that is a warning, not a measurement of this native Fourier network. Do not materialize huge truth tables or use approximate contraction to certify an exact zero count. Sorting interchangeable summands cannot be transplanted to distinct slot-specific bases without a new argument.

4. Instance-certified recursion. The literal native schedule divides its logarithmic size parameter by 131072, so machine-indexable graphs never get a positive production scale. Small forced tests check some transformed contractions against direct

counts; they prove neither general small-parameter validity nor speed. Separate transformation correctness from sufficient asymptotic shrinkage conditions. A proposed planner must check packing, coefficient/mode coverage, precision loss, reconstruction, child sizes, resource limits and termination. Try one layer with a priced exact fallback. Simply changing epsilon does not inherit the theorem. This route most directly attacks the same recursion's onset, and retains a substantial proof gap.

5. Nested hash thresholds and partial relations. Use an inner base and a larger outer base under the same hash. Retain relations with a bounded number of outer points and eliminate them later. Price outer-base setup, duplicates, storage, elimination and useful matrix rank. Preserve slot/point identities and coefficients modulo the odd subgroup order; binary cycle cancellation alone is insufficient. Earlier binary large-prime collector observations remain unqualified component evidence, with rho still ahead, and do not establish a gain here. Frobenius or negation orbits are another possible constant improvement, requiring rebuilt membership/enumeration/coverage arguments and an equally optimized rho.

Decision. Start with selector-domain witness search, then a width/propagation-aware backend. Keep checked adaptive recursion as a separate proof-and-planner track. Improvement over the current index-calculus baseline is useful even if rho remains ahead. A draft backlog card retains a possible lower-bit study; its protocol, driver, curves, checker and cost envelope are not ready. No experiment ran. Earlier S3 checkpoint upload/readback succeeded; native image/execution readiness remains a distinct prerequisite. The original frozen A/B panel, both challenge solves and independent audit stay open.

Identities, evidence and limits

Native source: `edfadbab5ac71471be7fd2971a9e50825307363f`. ECCp-131 short Weierstrass fixture:

```
p=1550031797834347859248576414813139942411
a=1399267573763578815877905235971153316710
b=1009296542191532464076260367525816293976
r=1550031797834347859219047037805205710577
cofactor=1
s=3102808443644027376164572183694178153449
```

ECC2K-130 is a separate binary problem; its full parameter source is linked in README.md. Its field degree 131 has no nontrivial intermediate subfield. Binary XOR/Frobenius ideas do not automatically transfer to the prime-field fixture. No new curve-panel input or timed run was generated. All historical measurements here are source reports, not reproduced results. No comprehensive novelty audit or independent revalidation of those sources was performed. No new algebraic theorem is claimed.

Sources checked. Bochum v0.2, sections 2–3 and cutoff footnote; Dudek et al., TensorOrder; Trimoska et al., SAT-based binary index calculus; pinned lead atlas; native relation and recursion code; historical binary bucket/cutset prototype. README.md links individual source and atlas entries, including negative SAT scaling and generic-oracle cost scopes. No canonical scoreboard, progress, curve, performance-gain or study result graph receives a new measured value or curve-transport edge. Editable diagram and PDF accompany this report.